

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

PRACTICAL-1:

Aim: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
venkatsaikrishna5@VSK-HP: ~$ mkdir OSSP
mkdir: OSSP: File exists
venkatsaikrishna5@VSK-HP: ~$ cd OSSP
venkatsaikrishna5@VSK-HP: ~/OSSP$ nano command_exec.c
venkatsaikrishna5@VSK-HP: ~/OSSP$ gcc command_exec.c -o command_exec
venkatsaikrishna5@VSK-HP: ~/OSSP$ ls
command_exec.c.save  command_exec.c
command_exec         command_exec.c.save
venkatsaikrishna5@VSK-HP: ~/OSSP$ ./command_exec
Enter a Linux command: uname

Parent Process
Parent PID: 3669

Child Process
Child PID: 3670
Parent PID: 3669
Linux
Child process completed.
venkatsaikrishna5@VSK-HP: ~/OSSP$ ./command_exec
Enter a Linux command: lscpu

Parent Process
Parent PID: 3680

Child Process
Child PID: 3681
Parent PID: 3680
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-1335
                        U
CPU family:             6
Model:                  186
Thread(s) per core:     2
```

```
venkatsaikrishna5@VSK-HP: ~  
Model: 186  
Thread(s) per core: 2  
Core(s) per socket: 6  
Socket(s): 1  
Stepping: 3  
BogoMIPS: 4991.99  
Flags: fpu vme de pse tsc msr pae mce cx8  
apic sep mtrr pge mca cmov pat ps  
e36 clflush mmx fxsr sse sse2 ss h  
t syscall nx pdpe1gb rdtscp lm con  
stant_tsc rep_good nopl xtopology  
tsc_reliable nonstop_tsc cpuid tsc  
_known_freq pni pclmulqdq vmx ssse  
3 fma cx16 pcid sse4_1 sse4_2 x2ap  
ic movbe popcnt tsc_deadline_timer  
aes xsave avx f16c rdrand hypervi  
sor lahf_lm abm 3dnowprefetch ssbd  
ibrs ibpb stibp ibrs_enhanced tpr  
_shadow ept vpid ept_ad fsgsbase t  
sc_adjust bmi1 avx2 smep bmi2 erms  
invpcid rdseed adx smap clflushop  
t clwb sha_ni xsaveopt xsavec xget  
bv1 xsaves avx_vnni vnm1 umip wait  
pkg gfni vaes vpclmulqdq rdpid mov  
diri movdir64b fsrm md_clear seria  
lize ibt flush_l1d arch_capabilities  
Virtualization features:  
Virtualization: VT-x  
Hypervisor vendor: Microsoft  
Virtualization type: full  
Caches (sum of all):  
L1d: 288 KiB (6 instances)  
L1i: 192 KiB (6 instances)  
L2: 7.5 MiB (6 instances)  
L3: 12 MiB (1 instance)  
NUMA:  
NUMA node(s): 1  
NUMA node0 CPU(s): 0-11  
Vulnerabilities:
```

```
venkatsaikrishna5@VSK-HP: ~  
Vulnerabilities:  
Gather data sampling: Not affected  
Ghostwrite: Not affected  
Indirect target selection: Not affected  
Itlb multihit: Not affected  
L1tf: Not affected  
Mds: Not affected  
Meltdown: Not affected  
Mmio stale data: Not affected  
Old microcode: Not affected  
Reg file data sampling: Mitigation; Clear Register File  
Retbleed: Mitigation; Enhanced IBRS  
Spec rstack overflow: Not affected  
Spec store bypass: Mitigation; Speculative Store Bypa  
ss disabled via prctl  
Spectre v1: Mitigation; usercopy/swapgs barrie  
rs and __user pointer sanitization  
Spectre v2: Mitigation; Enhanced / Automatic I  
BRS; IBPB conditional; PBRSE-eIBRS  
SW sequence; BHI BHI_DIS_S  
Srbds: Not affected  
Tsa: Not affected  
Tsx async abort: Not affected  
Vmscape: Not affected  
Child process completed.  
venkatsaikrishna5@VSK-HP: ~/OSSP$ ./command_exec  
Enter a Linux command: lsblk  
  
Parent Process  
Parent PID: 3693  
  
Child Process  
Child PID: 3694  
Parent PID: 3693  
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS  
sda 8:0 0 356.9M 1 disk  
sdb 8:16 0 159.4M 1 disk  
sdc 8:32 0 2G 0 disk [SWAP]  
sdd 8:48 0 1T 0 disk /mnt/wslg/distro  
/
```

```
venkatsaikrishna5@VSK-HP: ~  
Child process completed.  
venkatsaikrishna5@VSK-HP:~/OSSP$ ./command_exec  
Enter a Linux command: ps  
  
Parent Process  
Child Process  
Parent PID: 3704  
Child PID: 3705  
Parent PID: 3704  
PID TTY TIME CMD  
3578 pts/0 00:00:00 bash  
3704 pts/0 00:00:00 command_exec  
3705 pts/0 00:00:00 ps  
Child process completed.  
venkatsaikrishna5@VSK-HP:~/OSSP$ ./command_exec  
Enter a Linux command: top  
  
Parent Process  
Parent PID: 3715  
  
Child Process  
Child PID: 3716  
Parent PID: 3715  
top - 06:04:56 up 52 min, 1 user, load average: 0.00, 0.00, 0.00  
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 7796.6 total, 7001.0 free, 643.8 used, 311.8 buff/cache  
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7152.8 avail Mem  


| PID | USER     | PR | NI | VIRT  | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|-----|----------|----|----|-------|-------|-------|---|------|------|---------|-----------------|
| 1   | root     | 20 | 0  | 24144 | 14448 | 10604 | S | 0.0  | 0.2  | 0:01.75 | systemd         |
| 2   | root     | 20 | 0  | 3180  | 2204  | 2072  | S | 0.0  | 0.0  | 0:00.03 | init-systemd(Ub |
| 6   | root     | 20 | 0  | 3180  | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 49  | root     | 19 | -1 | 50400 | 16812 | 15524 | S | 0.0  | 0.2  | 0:00.39 | systemd-journal |
| 84  | systemd+ | 20 | 0  | 22540 | 14628 | 12012 | S | 0.0  | 0.2  | 0:00.14 | systemd-resolve |
| 93  | root     | 20 | 0  | 35360 | 12352 | 9268  | S | 0.0  | 0.2  | 0:02.23 | systemd-udev    |
| 113 | root     | 20 | 0  | 2888  | 1948  | 1820  | S | 0.0  | 0.0  | 0:00.06 | chronyd-starter |
| 114 | root     | 20 | 0  | 4472  | 3104  | 2860  | S | 0.0  | 0.0  | 0:00.02 | cron            |
| 116 | message+ | 20 | 0  | 8792  | 5492  | 4732  | S | 0.0  | 0.1  | 0:00.18 | dbus-daemon     |


```

```
venkatsaikrishna5@VSK-HP: ~  
top - 06:05:05 up 52 min, 1 user, load average: 0.00, 0.00, 0.00  
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 7796.6 total, 7012.0 free, 632.9 used, 311.8 buff/cache  
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7163.8 avail Mem  


| PID  | USER     | PR | NI | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|------|----------|----|----|--------|-------|-------|---|------|------|---------|-----------------|
| 3577 | root     | 20 | 0  | 3200   | 1256  | 1108  | S | 0.3  | 0.0  | 0:00.05 | Relay(3578)     |
| 1    | root     | 20 | 0  | 24144  | 14448 | 10604 | S | 0.0  | 0.2  | 0:01.75 | systemd         |
| 2    | root     | 20 | 0  | 3180   | 2204  | 2072  | S | 0.0  | 0.0  | 0:00.03 | init-systemd(Ub |
| 6    | root     | 20 | 0  | 3180   | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 49   | root     | 19 | -1 | 50400  | 16812 | 15524 | S | 0.0  | 0.2  | 0:00.39 | systemd-journal |
| 84   | systemd+ | 20 | 0  | 22540  | 14628 | 12012 | S | 0.0  | 0.2  | 0:00.14 | systemd-resolve |
| 93   | root     | 20 | 0  | 35360  | 12352 | 9268  | S | 0.0  | 0.2  | 0:02.23 | systemd-udev    |
| 113  | root     | 20 | 0  | 2888   | 1948  | 1820  | S | 0.0  | 0.0  | 0:00.06 | chronyd-starter |
| 114  | root     | 20 | 0  | 4472   | 3104  | 2860  | S | 0.0  | 0.0  | 0:00.02 | cron            |
| 116  | message+ | 20 | 0  | 8792   | 5492  | 4732  | S | 0.0  | 0.1  | 0:00.18 | dbus-daemon     |
| 159  | root     | 20 | 0  | 44092  | 29780 | 14648 | S | 0.0  | 0.4  | 0:00.40 | networkd-dispat |
| 163  | root     | 20 | 0  | 18492  | 9528  | 8224  | S | 0.0  | 0.1  | 0:00.21 | systemd-logind  |
| 177  | syslog   | 20 | 0  | 220548 | 5892  | 4624  | S | 0.0  | 0.1  | 0:00.12 | rsyslogd        |
| 216  | root     | 20 | 0  | 5492   | 2712  | 2528  | S | 0.0  | 0.0  | 0:00.01 | agetty          |
| 217  | _chrony  | 20 | 0  | 20420  | 6384  | 5608  | S | 0.0  | 0.1  | 0:00.14 | chronyd         |
| 232  | _chrony  | 20 | 0  | 12092  | 2448  | 1804  | S | 0.0  | 0.0  | 0:00.00 | chronyd         |
| 246  | root     | 20 | 0  | 123456 | 32620 | 18004 | S | 0.0  | 0.4  | 0:00.34 | unattended-upgr |
| 318  | root     | 20 | 0  | 8648   | 5516  | 4672  | S | 0.0  | 0.1  | 0:00.02 | login           |
| 384  | venkats+ | 20 | 0  | 21936  | 11380 | 9176  | S | 0.0  | 0.1  | 0:00.37 | systemd         |
| 386  | venkats+ | 20 | 0  | 22912  | 4112  | 2112  | S | 0.0  | 0.1  | 0:00.00 | (sd-pam)        |
| 412  | venkats+ | 20 | 0  | 6136   | 5416  | 3864  | S | 0.0  | 0.1  | 0:00.04 | bash            |
| 828  | polkitd  | 20 | 0  | 306444 | 8444  | 7628  | S | 0.0  | 0.1  | 0:00.21 | polkitd         |
| 3576 | root     | 20 | 0  | 3184   | 1116  | 980   | S | 0.0  | 0.0  | 0:00.00 | SessionLeader   |
| 3578 | venkats+ | 20 | 0  | 6268   | 5544  | 3872  | S | 0.0  | 0.1  | 0:00.11 | bash            |
| 3715 | venkats+ | 20 | 0  | 2768   | 1820  | 1720  | S | 0.0  | 0.0  | 0:00.00 | command_exec    |
| 3716 | venkats+ | 20 | 0  | 8172   | 6052  | 3948  | R | 0.0  | 0.1  | 0:00.03 | top             |


```

Code:

```
venkatsaikrishna5@VSK-HP: ~  
GNU nano 8.7.1 command_exec.c  
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <sys/wait.h>  
#include <string.h>  
  
int main() {  
    char command[100];  
  
    printf("Enter a Linux command: ");  
    fgets(command, sizeof(command), stdin);  
  
    // Remove newline character  
    command[strcspn(command, "\n")] = '\0';  
  
    pid_t pid = fork();  
  
    if (pid < 0) {  
        printf("Fork failed!\n");  
        return 1;  
    }  
    else if (pid == 0) {  
        // Child Process  
        printf("\nChild Process\n");  
        printf("Child PID: %d\n", getpid());  
        printf("Parent PID: %d\n", getppid());  
  
        execlp(command, command, NULL);  
  
        // Executes only if exec fails  
        printf("Invalid command!\n");  
        exit(1);  
    }  
    else {  
        // Parent Process  
        printf("\nParent Process\n");  
  
        ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  
        ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_/ Go To Line M-E Redo      M-G Copy
```

```
venkatsaikrishna5@VSK-HP: ~  
GNU nano 8.7.1 command_exec.c  
    fgets(command, sizeof(command), stdin);  
  
    // Remove newline character  
    command[strcspn(command, "\n")] = '\0';  
  
    pid_t pid = fork();  
  
    if (pid < 0) {  
        printf("Fork failed!\n");  
        return 1;  
    }  
    else if (pid == 0) {  
        // Child Process  
        printf("\nChild Process\n");  
        printf("Child PID: %d\n", getpid());  
        printf("Parent PID: %d\n", getppid());  
  
        execlp(command, command, NULL);  
  
        // Executes only if exec fails  
        printf("Invalid command!\n");  
        exit(1);  
    }  
    else {  
        // Parent Process  
        printf("\nParent Process\n");  
        printf("Parent PID: %d\n", getpid());  
  
        wait(NULL);  
  
        printf("Child process completed.\n");  
    }  
  
    return 0;  
}
```